

Proposed Solution Template

ProjectDesignPhaseReport

Date	1 OCTOBER 2026	Maximum Marks	2 Marks
Team ID	SWTID-2026-6919	Project Name	Script-Controlled ACL - Restrict Record Access Based on Field Value

Proposed Solution Details

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	In ServiceNow instances, sensitive IT and business data records are often accessible or editable beyond allowable workflow stages or specific field conditions. Out-of-the-box (OOTB) declarative Access Control Lists (ACLs) rely primarily on static roles and basic condition builders, which cannot dynamically evaluate complex runtime logic (e.g., restricting field updates when a ticket reaches a 'Closed' or 'In Progress' state, or enforcing dual-attribute security conditions). Relying solely on client-side form controls (UI Policies / Client Scripts) leaves backend REST/SOAP APIs and reporting tools exposed to unauthorized data modifications.
2.	Idea / Solution description	Implementation of a server-side Script-Controlled Access Control List (ACL) Rule (sys_security_acl) utilizing server-side JavaScript evaluation. The script dynamically evaluates runtime record field values (current.<field_name>) against current user attributes (gs.getUser(), session roles, group memberships) directly on the server before granting read, write, create, or delete permissions. This ensures universal security enforcement across all data access interfaces including Web UI, Workspace, REST APIs, and Reporting engines.
3.	Novelty / Uniqueness	• Universal API & UI Hardening: Unlike UI-level masking, script-controlled ACLs enforce access control at the database transaction layer, eliminating security loopholes via REST/SOAP APIs or background scripts. • Dynamic Runtime Evaluation: Evaluates multi-variable dynamic conditions without causing 'role explosion' (creating excessive custom roles for every state combination). • Performance-Optimized Design: Designed with lightweight JavaScript execution without heavy GlideRecord loops to maintain peak database query performance.
4.	Social Impact / Customer Satisfaction	• Data Governance & Compliance Assurance: Provides security managers and compliance auditors with guaranteed data protection against unauthorized field tampering. • Seamless User Experience: Dynamically grants or restricts access based on operational context, eliminating user confusion caused by unnecessary error messages or rigid static roles. • Regulatory Preparedness: Satisfies strict enterprise security compliance standards (e.g., ISO 27001, SOC 2, OWASP data privacy mandates).

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5.	Business Model (Revenue Model)	• Enterprise Cost Avoidance: Mitigates financial risks associated with security breaches, data leaks, and regulatory compliance penalties. • Reduced Maintenance Overhead: Replaces dozens of complex client-side scripts and custom roles with streamlined, maintainable server-side security logic. • Asset Reusability: Packaged as a reusable Scoped Update Set that can be quickly deployed across enterprise ServiceNow instances.
6.	Scalability of the Solution	• High Performance & Instance Scalability: Executes within milliseconds at the database query layer, ensuring smooth performance even across millions of records. • Cross-Table Applicability: Easily adaptable across any ServiceNow table (Incident, Change Request, HR Service Delivery, Custom Scoped Application tables). • Centralized Management: Can be extended using Script Includes for centralized security policy management across the entire enterprise platform.